

Project Technical Breakdown

1. Executive Summary

NeuroFinance.AI is an end-to-end Financial Decision Intelligence platform. The system aggregates historical credit profiles, trains a deep learning Feature Tokenizer and Transformer (FT-Transformer) model, explains complex credit predictions using Shapley Additive exPlanations (SHAP), and provides interactive conversational assistance using a self-healing Retrieval-Augmented Generation (RAG) banking policy circular lookup system.

Recently, a real-time CNN Document Intelligence module was added, enabling automatic borrower verification by classifying uploaded files (payslips, bank statements) via a 3-layer Convolutional Neural Network, extracting text through OCR, and simulating credit risk impact on the fly.

2. Tabular Credit Scoring Model (FT-Transformer)

Unlike standard gradient boosting trees (XGBoost/LightGBM), NeuroFinance.AI implements a Feature Tokenizer + Transformer (FT-Transformer) architecture built in PyTorch. This deep learning model maps numerical and categorical features to high-dimensional token embeddings, processes token interactions through multi-head self-attention, and runs predictions through a binary classification feed-forward network:

- Feature Tokenization: Numerical features are scaled and multiplied by parameter weights. Categorical features are mapped through lookup embedding matrices.
- Multi-Head Self-Attention: Learns complex feature interactions directly in the latent space.
- Class Weight Balancing: Cross-entropy loss is weighted to counteract the 8% default imbalance.

Model Training Evaluation Metrics:

Metric	Value
ROC-AUC Score	0.6685
PR-AUC Score	0.1480
Feature Count	261 Features

3. SHAP Explainability Service

To satisfy banking explainability requirements, the system implements a SHAP explanation service. It uses a background baseline dataset of preprocessed records to compute Shapley values for each borrower. This isolates individual features (like external bureau ratings, monthly net income, and age) to show how much they push the risk probability up (red) or down (green).

Fault-Tolerant Attributions: If the base SHAP package throws platform-specific errors on CPU machines, the service catches it and falls back to a PyTorch gradient-based attribution wrapper (Input-Times-Gradient). This ensures explaining queries are 100% reliable.

4. CNN Document Intelligence & OCR Extraction

The Document Intelligence module automates verification by classifying uploaded document scans and extracting relevant financial details. It consists of three distinct steps:

- 1. CNN Layout Classification: Uses a PyTorch 3-layer Convolutional Neural Network (Conv2D) to identify if the document is a PAYSリップ, BANK_STATEMENT, LOAN_STATEMENT, or CREDIT_CARD_STATEMENT.
- 2. OCR Text Parsing: Scans the document content (with fallback to local transcripts for robust performance).
- 3. Structured Extraction & Validation: Uses line-bounded regular expressions to extract metrics like monthly income and delinquent flags. It runs validation checks to catch low confidence or missing values.

CNN Document Classifier Performance:

Document Type	Precision	Recall	F1-Score
PAYSリップ	1.0000	1.0000	1.0000
BANK STATEMENT	0.5385	1.0000	0.7000
CREDIT CARD	1.0000	1.0000	1.0000
WEIGHTED AVG	0.6581	0.7778	0.7000

5. NeuroBot Conversational Agent Orchestration

NeuroBot serves as the conversational orchestrator, connecting users directly to core system features through natural language queries. NeuroBot routes intents dynamically to specialized sub-agents:

- Risk Agent: Invokes the PyTorch credit network and formatting attributions.
- RAG Agent: Performs persistent vector lookup in ChromaDB for policy checks.
- Finance Agent: Performs calculations for loan principal, rate, and tenure EMI.
- Document Agent: Runs CNN verification and merges extracted metrics into default predictions.

This unified agentic loop connects document uploading and credit-risk modeling dynamically, allowing underwriters to analyze document impacts on risk scores with a single conversational query.

System Architecture Flow:

User Uploads Document -> CNN Classification (Document Type) -> OCR Processing (Raw Text) -> Structured Feature Extraction (Payslip Monthly Income) -> Validation Alert Checks -> Feature Overwrite (Updates Baseline Profile) -> FT-Transformer Scoring -> SHAP Explainability -> Conversational Chatbot Report Output.